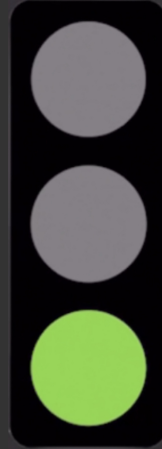


binary representation

CSCI
134

what do these mean?



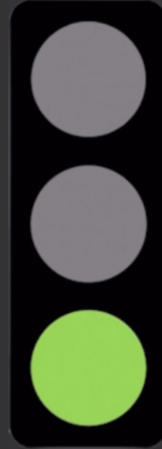
what do these mean?



"stop"



"slow down"



"go"

patterns of lights can
communicate messages



"stop"



"slow down"

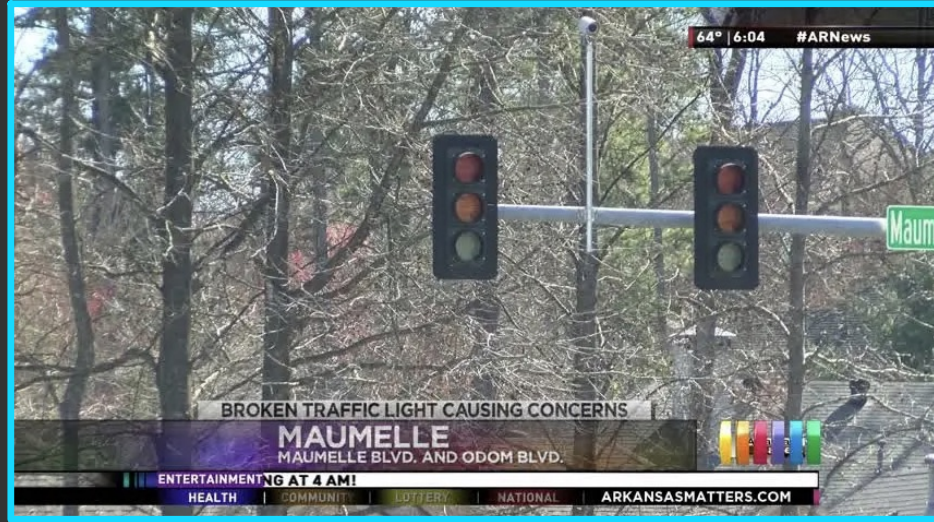


"go"

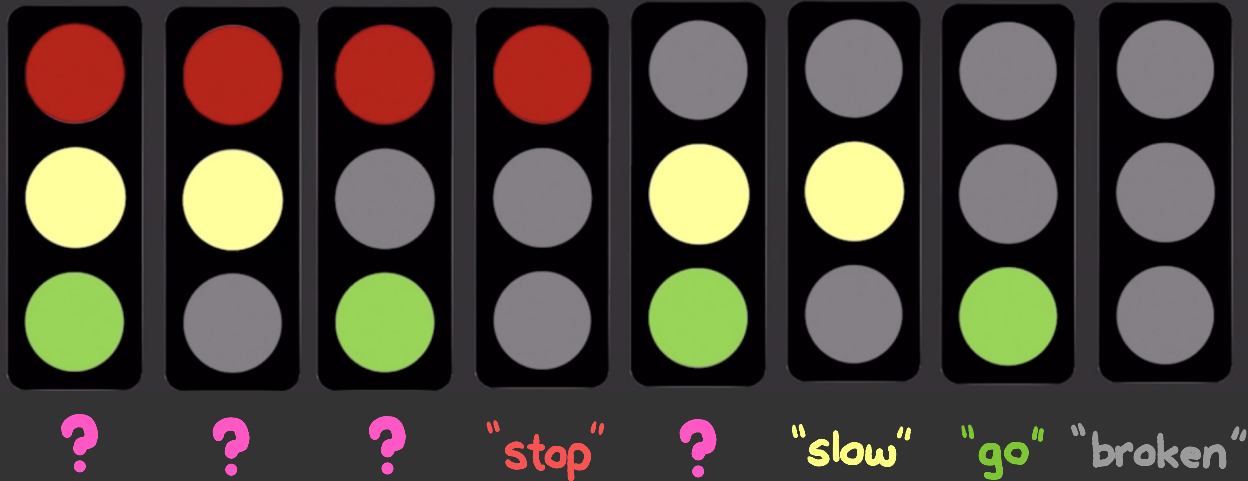
patterns of lights can
communicate messages



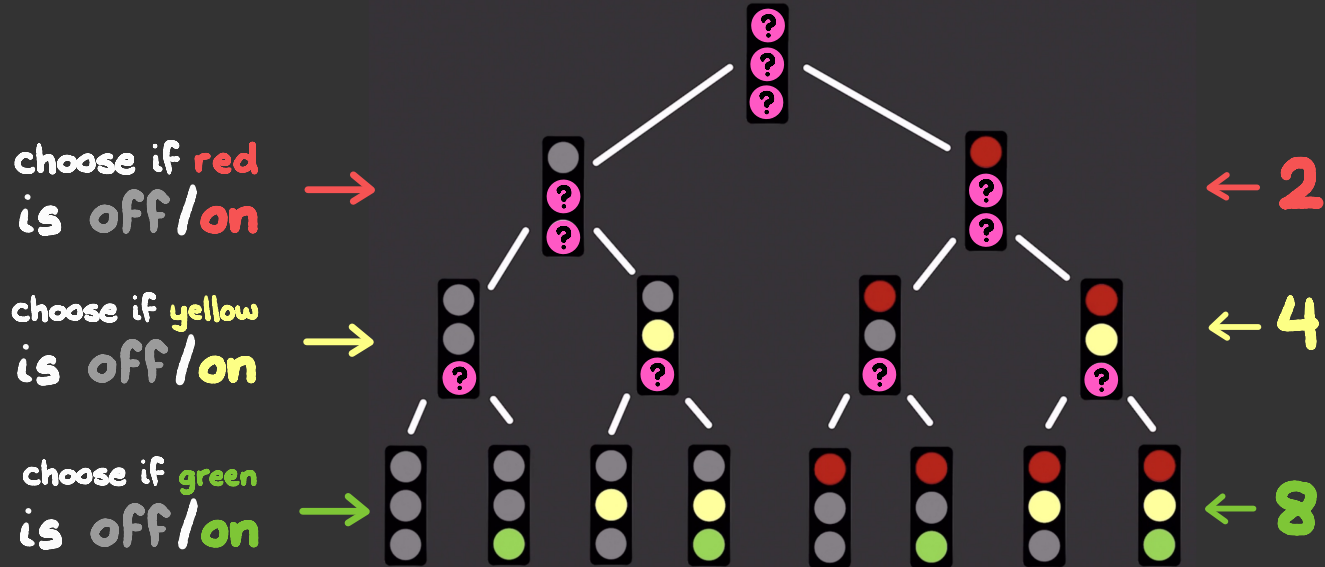
"broken"



3 lights have the capacity
to communicate 8 messages

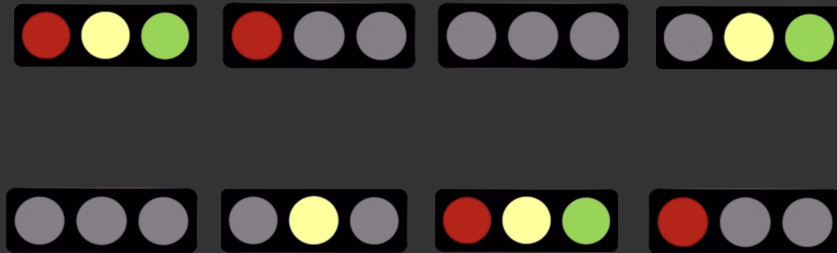
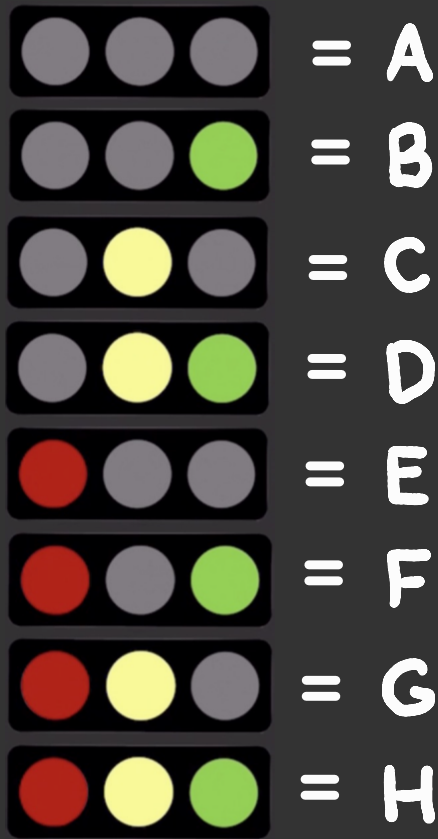


3 lights have the capacity
to communicate 8 messages



	= A
	= B
	= C
	= D
	= E
	= F
	= G
	= H

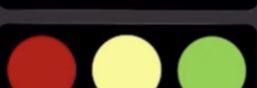
for instance, we
could use traffic
lights to encode
the first **8** letters
of the english
alphabet



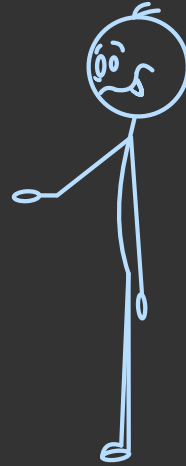
	=	A
	=	B
	=	C
	=	D
	=	E
	=	F
	=	G
	=	H

H	E	A	D
A	C	H	E



	=	A
	=	B
	=	C
	=	D
	=	E
	=	F
	=	G
	=	H

ADVIL?



how many lights would we need to represent the **entire** english alphabet?

 = A

 = B

 = C

 = D

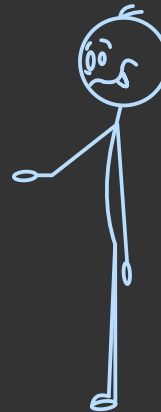
 = E

 = F

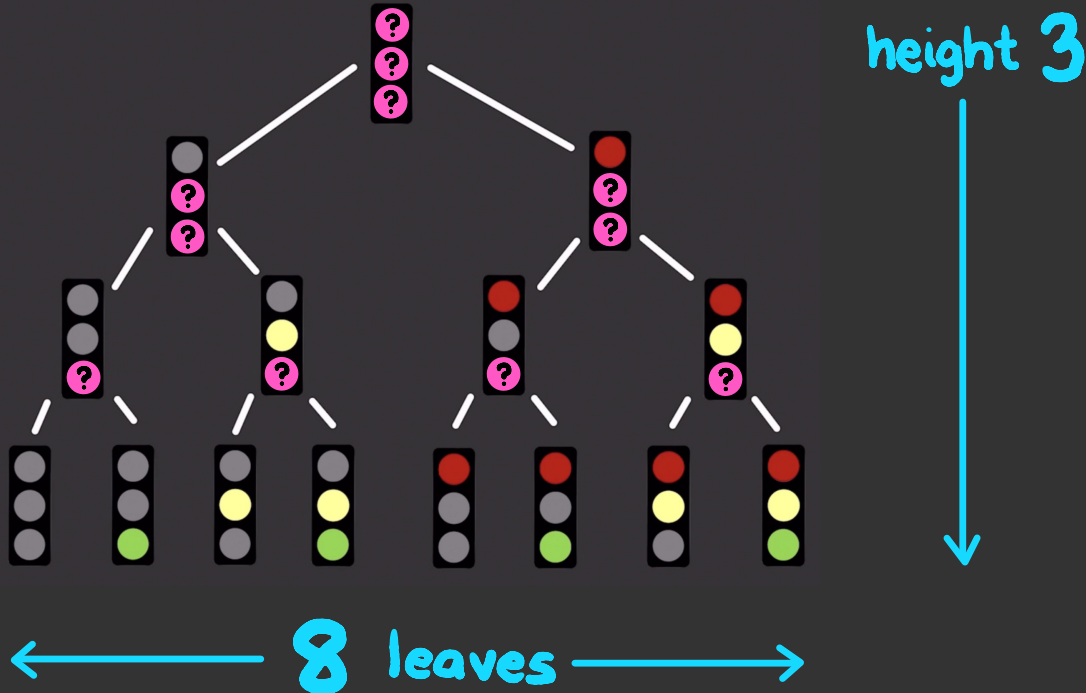
 = G

 = H

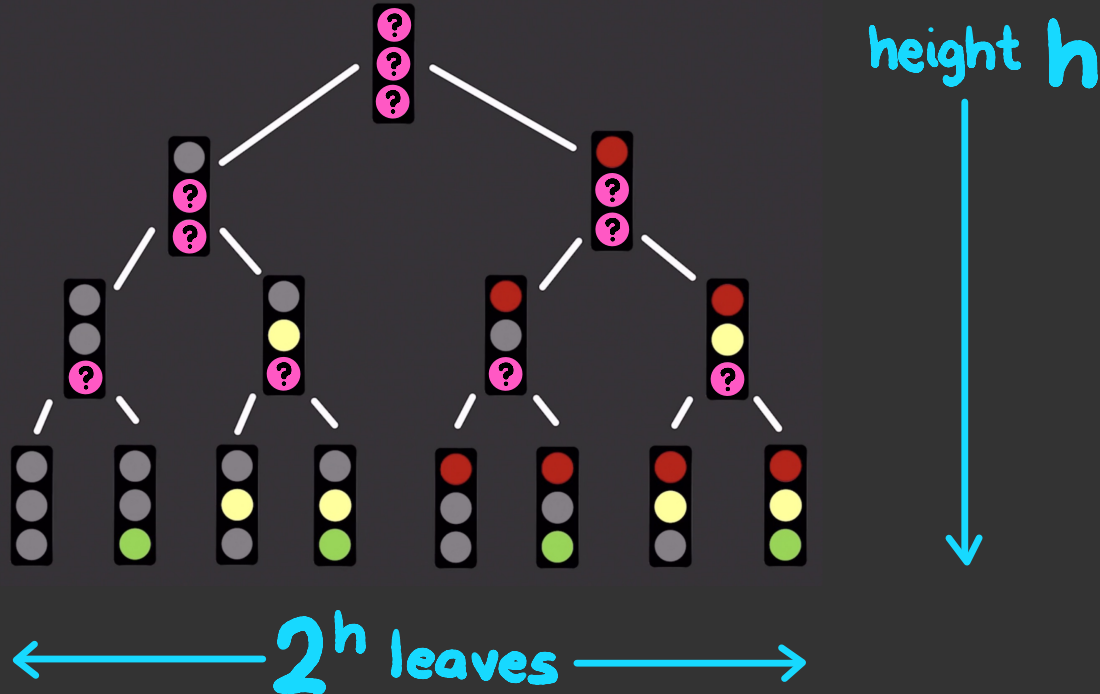
ADVIL?



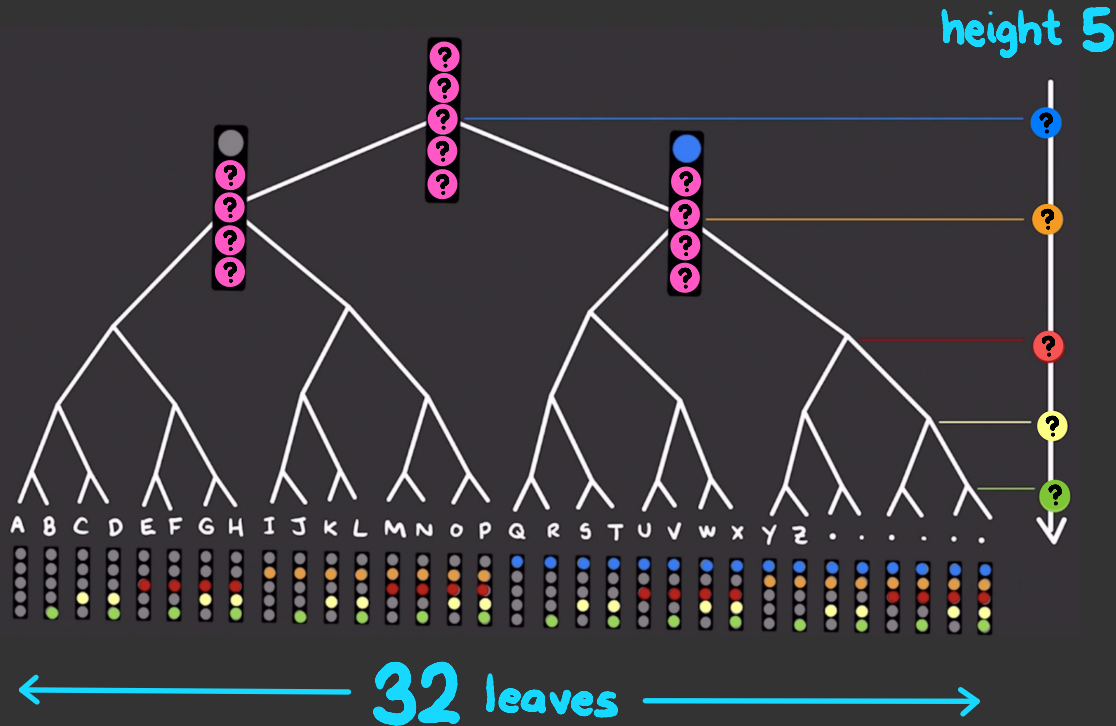
how many lights would we need to represent the **entire** english alphabet?



how many lights would we need to represent the **entire** english alphabet?



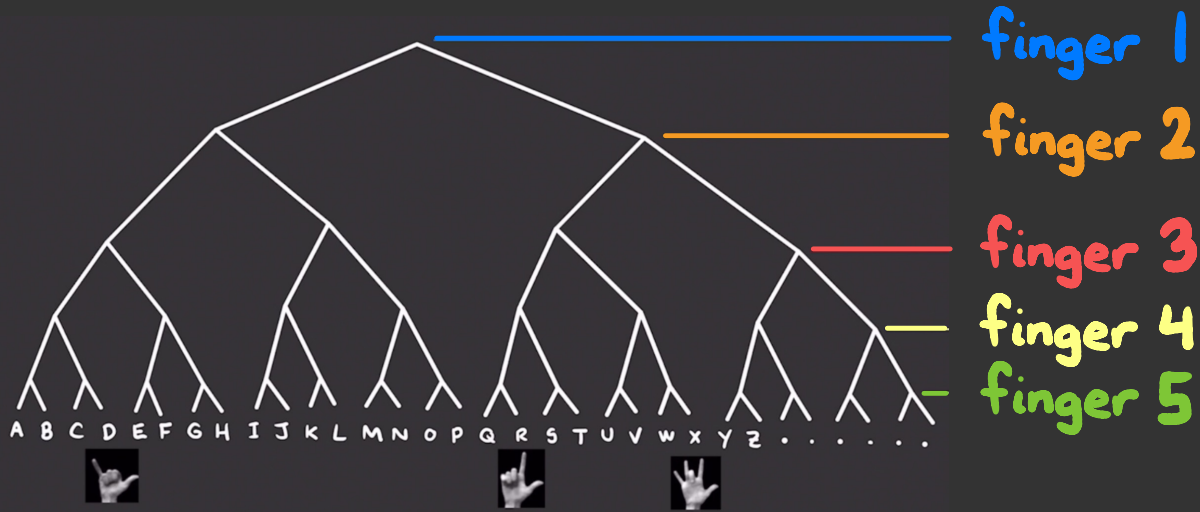
how many lights would we need to represent the **entire** english alphabet?



how many letters can we
represent with one hand?



how many letters can we
represent with one hand?



$2^h = 32$

if we have k on-off switches,
then we can represent 2^k values

e.g. light bulbs, fingers



if we have k on-off switches,
then we can represent 2^k values

e.g. letters, integers,
traffic messages



if we have k on-off switches,
then we can represent 2^k values

if we want to represent n values,
how many switches do we need?

???

if we have k on-off switches,
then we can represent 2^k values

if we want to represent n values,
how many switches do we need?

$$\lceil \log_2 n \rceil$$

bitstrings

a standard way to represent on/off switches

each zero or one
is referred to
as a **bit**
(**b**inary **d**igit)

bitstrings of
length **k**
can encode
 2^k values

	<u>bitstring</u>
	1 0 0 0 1
	1 0 1 1 1
	0 0 0 1 1

	<u>bitstring</u>
	1 0 0
	0 1 0
	0 0 1